

- **Available bit rate (ABR) ATM network service.** With the Internet offering so-called best-effort service, ATM's ABR might best be characterized as being a slightly-better-than-best-effort service. As with the Internet service model, cells may be lost under ABR service. Unlike in the Internet, however, cells cannot be reordered (although they may be lost), and a minimum cell transmission rate (MCR) is guaranteed to a connection using ABR service. If the network has enough free resources at a given time, a sender may also be able to send cells successfully at a higher rate than the MCR. Additionally, as we saw in Section 3.6, ATM ABR service can provide feedback to the sender (in terms of a congestion notification bit, or an explicit rate at which to send) that controls how the sender adjusts its rate between the MCR and an allowable peak cell rate.

4.2 Virtual Circuit and Datagram Networks

Recall from Chapter 3 that a transport layer can offer applications connectionless service or connection-oriented service between two processes. For example, the Internet's transport layer provides each application a choice between two services: UDP, a connectionless service; or TCP, a connection-oriented service. In a similar manner, a network layer can provide connectionless service or connection service between two hosts. Network-layer connection and connectionless services in many ways parallel transport-layer connection-oriented and connectionless services. For example, a network-layer connection service begins with handshaking between the source and destination hosts; and a network-layer connectionless service does not have any handshaking preliminaries.

Although the network-layer connection and connectionless services have some parallels with transport-layer connection-oriented and connectionless services, there are crucial differences:

- In the network layer, these services are host-to-host services provided by the network layer for the transport layer. In the transport layer these services are process-to-process services provided by the transport layer for the application layer.
- In all major computer network architectures to date (Internet, ATM, frame relay, and so on), the network layer provides either a host-to-host connectionless service or a host-to-host connection service, but not both. Computer networks that provide only a connection service at the network layer are called **virtual-circuit (VC) networks**; computer networks that provide only a connectionless service at the network layer are called **datagram networks**.
- The implementations of connection-oriented service in the transport layer and the connection service in the network layer are fundamentally different. We saw in the previous chapter that the transport-layer connection-oriented service is

implemented at the edge of the network in the end systems; we'll see shortly that the network-layer connection service is implemented in the routers in the network core as well as in the end systems.

Virtual-circuit and datagram networks are two fundamental classes of computer networks. They use very different information in making their forwarding decisions. Let's now take a closer look at their implementations.

4.2.1 Virtual-Circuit Networks

While the Internet is a datagram network, many alternative network architectures—including those of ATM and frame relay—are virtual-circuit networks and, therefore, use connections at the network layer. These network-layer connections are called **virtual circuits (VCs)**. Let's now consider how a VC service can be implemented in a computer network.

A VC consists of (1) a path (that is, a series of links and routers) between the source and destination hosts, (2) VC numbers, one number for each link along the path, and (3) entries in the forwarding table in each router along the path. A packet belonging to a virtual circuit will carry a VC number in its header. Because a virtual circuit may have a different VC number on each link, each intervening router must replace the VC number of each traversing packet with a new VC number. The new VC number is obtained from the forwarding table.

To illustrate the concept, consider the network shown in Figure 4.3. The numbers next to the links of R1 in Figure 4.3 are the link interface numbers. Suppose now that Host A requests that the network establish a VC between itself and Host B. Suppose also that the network chooses the path A-R1-R2-B and assigns VC numbers 12, 22, and 32 to the three links in this path for this virtual circuit. In this case, when a packet in this VC leaves Host A, the value in the VC number field in the packet header is 12; when it leaves R1, the value is 22; and when it leaves R2, the value is 32.

How does the router determine the replacement VC number for a packet traversing the router? For a VC network, each router's forwarding table includes VC

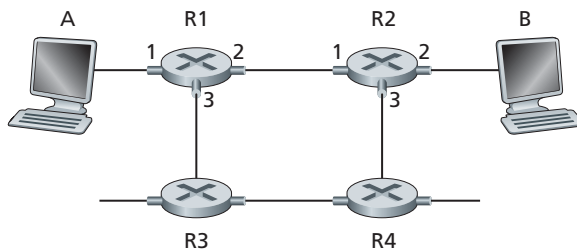


Figure 4.3 ♦ A simple virtual circuit network